particles. As one primary example, subsequent work by Strominger and Witten showed that the masses of the particles in each family depend upon—hang on, this is a bit tricky—the way in which the boundaries of the various multidimensional holes in the Calabi-Yau shape intersect and overlap with one another. It's hard to visualize, but the idea is that as strings vibrate through the extra curled-up dimensions, the precise arrangement of the various holes and the way in which the Calabi-Yau shape folds around them has a direct impact on the possible resonant patterns of vibration. Although the details get difficult to follow and are really not all that essential, what is important is that, as with the number of families, string theory can provide us with a framework for answering questions—such as why the electron and other particles have the masses they do—on which previous theories are completely silent. Once again, though, carrying through with such calculations requires that we know which Calabi-Yau space to take for the extra dimensions.

The preceding discussion gives some idea of how string theory may one day explain the properties of the matter particles recorded in Table 1.1. String theorists believe that a similar story will one day also explain the properties of the messenger particles of the fundamental forces, listed in Table 1.2. That is, as strings twist and vibrate while meandering through the extended and curled-up dimensions, a small subset of their vast oscillatory repertoire consists of vibrations with spin equal to 1 or 2. These are the candidate force-carrying string-vibrational states. Regardless of the shape of the Calabi-Yau space, there is always one vibrational pattern that is massless and has spin-2; we identify this pattern as the graviton. The precise list of spin-1 messenger particles—their number, the strength of the force they convey, the gauge symmetries they respect—though, does depend crucially on the precise geometrical form of the curled-up dimensions. And so, once again, we come to realize that string theory provides a framework for explaining the observed messenger-particle content of our universe, that is, for explaining the properties of the fundamental forces, but that without knowing exactly which Calabi-Yau shape the extra dimensions are curled into, we cannot make any definitive predictions or postdictions (beyond Witten's remark regarding the postdiction of gravity).